

A Multimodal Vision Model for Automated Industrial Safety and Workforce Monitoring

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Abstract—Industrial workplaces continue to face persistent safety challenges that manual supervision alone cannot adequately address. This paper describes the design and implementation of a real-time safety monitoring system that combines computer vision, deep learning, and cloud-based analytics to automate key aspects of workplace hazard detection and access control. Rather than relying on periodic human inspection, the system continuously processes video feeds from standard cameras to identify unsafe conditions as they occur.

The system covers several practical safety scenarios: checking whether workers are wearing required protective gear such as helmets and safety vests, detecting falls, identifying unauthorized individuals in restricted zones, reading vehicle license plates, and screening for alcohol impairment before granting area access. A separate facial recognition module handles attendance verification, ensuring that only credentialed personnel are logged as present. All detected events are stored in a centralized cloud database along with timestamps and image snapshots, which feed into a supervisor dashboard for live monitoring and historical review.

Testing across varied lighting and environmental conditions showed that each subsystem performs reliably without significant degradation. The architecture is deliberately modular, making it straightforward to add new detection capabilities without restructuring the existing pipeline. Beyond its technical contributions, the work demonstrates that automated, vision-based monitoring can meaningfully reduce the burden on human supervisors while improving response times to potential incidents. The approach is particularly suited to industrial settings where safety compliance is both critical and difficult to enforce consistently at scale.

Index Terms—Alcohol analysis, ANPR, computer vision, face recognition, PPE detection.

I. INTRODUCTION

The nature of industrial work has changed considerably over the past decade. Automation has taken over many routine tasks, production lines have grown more interconnected, and the organizational logic of Industry 4.0 has pushed factories toward tighter integration of machines, data, and human workers. These shifts have brought genuine productivity gains, but they have also introduced new layers of complexity that safety management has struggled to keep pace with.

What has not changed, however, is where most accidents come from. Workers skip or misuse protective equipment, unauthorized individuals enter restricted areas, high-risk zones go unmonitored for stretches of time, and impaired workers occasionally make it past manual checkpoints. These are not novel failure modes—they are well-documented, recurring patterns that existing supervision practices have never fully solved [1][10]. The honest reason is that manual observation simply does not scale. A human supervisor can watch one area at a time, grows fatigued, and cannot generate a timestamped evidence trail without extra effort. In facilities that run around the clock across multiple zones, these limitations are structural, not incidental.

Computer vision and deep learning have opened a practical path forward. Over the last five years, the research focus has shifted noticeably from narrow, single-purpose detectors toward broader monitoring frameworks that handle several safety tasks within one architecture [2][10]. PPE compliance detection has matured considerably during this period, with convolutional network approaches proving effective at identifying missing helmets and vests under real operating conditions [1][8]. Researchers have also explored user-centered designs that make these systems easier to deploy and interpret in practice [4].

Parallel progress has occurred in environment-aware monitoring. Systems built for human-robot collaboration have demonstrated that visual safety checks and operational quality monitoring can coexist within a single pipeline [3], and computer vision-based ergonomic assessment tools have been validated directly on factory floors rather than in controlled lab settings [6]. The broader Safety 4.0 agenda has further accelerated adoption, with manufacturing facilities treating computer vision as a core infrastructure layer rather than an experimental add-on [2].

Activity recognition and anomaly detection have followed a similar trajectory. Deep learning models applied to surveillance footage have shown the ability to flag suspicious be-

havior with meaningful accuracy [5], and weakly-supervised approaches using I3D convolution networks have performed well on anomaly identification without requiring exhaustively labeled training data [7]. Vehicle access control has also received attention, with automatic number plate recognition becoming a standard tool for managing entry in industrial compounds [9].

Taken together, these developments suggest that the individual components of a comprehensive safety system now exist and have been separately validated. What remains less explored is how they perform when brought together into a single, continuously operating platform—one that handles PPE detection, intrusion monitoring, fall detection, alcohol screening, facial recognition, and vehicle authorization in parallel, logs every event with image evidence, and feeds structured data to a dashboard that supervisors can actually use [10]. That integration challenge, and the system we built to address it, is what this paper describes.

II. LITERATURE SURVEY

PPE detection has probably attracted more research attention than any other single topic in industrial computer vision over the past several years. The systematic review by Vukicevic et al. [1] gives a clear picture of where the field stands: models like YOLOv5, YOLOv8, and Faster R-CNN now achieve strong detection accuracy for helmets, vests, gloves, and full-body protective wear under benchmark conditions. The harder problem, as their review makes clear, is getting these systems to hold up in actual factories, where lighting shifts throughout the day, workers move unpredictably, and the same type of safety gear can look very different across manufacturers and individuals.

Yousif et al. [2] situate PPE detection within the broader Safety 4.0 agenda, arguing that computer vision should function as a persistent safety observer built into smart manufacturing infrastructure rather than a bolt-on tool. Their work shows that continuous automated compliance checking is both technically feasible and operationally useful, particularly in facilities too large for supervisors to cover on foot. On the deployment side, Trujillo-López et al. [4] raise a point that purely accuracy-focused research tends to undervalue: a detection system is only as useful as workers’ ability to understand and respond to its alerts. Their user-centered design study found that integrating visual warnings directly into operational workflows, rather than routing them through a separate monitoring interface, significantly improved how workers actually engaged with PPE reminders. Medina Villagomez et al. [8] address a different practical constraint, showing that lightweight CNN architectures can run PPE recognition on edge devices without heavy computing infrastructure. This matters because most discussions of industrial AI implicitly assume enterprise-scale hardware budgets, which excludes a large share of real manufacturing environments.

Beyond PPE, factories introduce hazards that arise specifically from human-machine interaction. Kozamernik et al. [3] focus on cooperative work cells where humans and robots

share physical space, demonstrating that vision-based proximity tracking can trigger protective shutdowns before a collision occurs. Their real-world validation is worth noting because much work in this space stops at simulation; theirs does not.

Behavioral anomalies represent another category of risk that PPE detection alone cannot catch. Wani et al. [5] develop a surveillance pipeline capable of flagging unusual activity patterns in real-time video, covering behaviors like loitering, aggressive motion, and access irregularities. Agostinelli et al. [6] take a complementary angle, using pose estimation to identify ergonomic risk factors such as improper lifting posture and repetitive strain movements before they result in injury. Both approaches push safety monitoring into behavior analysis territory, which is a meaningful extension beyond checking whether the right equipment is worn.

For detecting threats that are harder to define in advance, Soltani Nejad and Haque [7] propose a weakly supervised anomaly detection model built on two-stream I3D networks that process both RGB frames and optical flow. The weakly supervised framing is practically significant because exhaustively labeling anomalous events for training is expensive and often incomplete in real deployments. Their approach handles unforeseen threats such as unusual movement trajectories and unauthorized nighttime activity without requiring a predefined catalog of what “suspicious” looks like.

Vehicle access control is frequently treated as a separate security problem, but in logistics-heavy industrial environments it belongs in the same monitoring conversation. Ditta et al. [9] present a deep learning ANPR system that reads license plates reliably across varied camera angles and outdoor lighting conditions. Although developed for commercial parking, the underlying mechanism transfers directly to factory entry points, where vehicles need to be authenticated against an access list and logged with timestamps. Unauthorized vehicle entry is a surprisingly common security gap in industrial facilities, and ANPR offers a low-friction way to close it.

Ozobu et al. [10] take a step back from individual detection tasks and ask what a genuinely comprehensive safety framework would need to include. Their answer involves fusing environmental sensor data, including alcohol detection, temperature readings, and gas monitoring, with real-time video analysis and predictive analytics. The key finding is that combining these signals produces a more reliable picture of workspace risk than any single modality can provide on its own, and that early warning becomes possible when the system can cross-reference visual observations with sensor context.

The body of work reviewed here covers the main building blocks of industrial safety monitoring: PPE compliance [1][2][4][8], human-machine environment awareness [3], activity and behavioral risk detection [5][6], spatiotemporal anomaly modeling [7], vehicle authorization [9], and multi-modal sensor integration [10]. What these studies collectively suggest is that the components exist and have been individually validated. The less-explored question is whether they can be made to work together reliably within a single operational system, and that is what the present work attempts to address.

III. METHODOLOGY

The system is built around four loosely coupled layers that handle data acquisition, AI inference, output validation, and persistent storage. Keeping these layers separate was a deliberate architectural choice: it makes individual components easier to test, replace, or extend without touching the rest of the pipeline.

The monitoring interface is developed using Next.js 15 with TypeScript and Tailwind CSS. Live video is streamed through the browser's `getUserMedia` API and rendered via standard HTML video elements. A hidden canvas element runs in parallel, periodically pulling still frames from the video stream and encoding them as Base64 strings. Each of the five monitoring modules operates its own independent capture loop using `setInterval`, so a slowdown in one module does not stall the others.

Captured frames are forwarded to the server through secure Next.js server actions. On the server side, the Base64 image is decoded, the appropriate trained model is loaded for the given task, a forward pass is executed, and the result is returned as structured JSON. Rather than relying on a single general-purpose vision model with dynamic prompting, each monitoring task uses a dedicated model trained on task-specific annotated data. The training pipeline followed standard practice: dataset collection, preprocessing and augmentation, training using modern deep learning architectures, and export to an optimized inference format. All model outputs are validated against Zod-defined schemas before being passed downstream, which keeps output structure consistent even when confidence scores vary.

The system implements five monitoring modules, each following the same internal workflow: capture, transmit, infer, validate, store, and notify. PPE compliance detection processes incoming frames to check for helmets and reflective vests, generating a violation log entry and frame snapshot whenever required equipment is absent. Number plate recognition extracts plate text and cross-references it against a stored authorization list, classifying each vehicle as known, unknown, or restricted and logging the result with a timestamp and image. Intrusion detection monitors for unregistered individuals during configured operational hours, raising alerts when unauthorized presence is identified in restricted zones. Alcohol detection works through an MQ-3 breath sensor: readings above the configured threshold automatically trigger a snapshot capture, event log entry, and cloud storage upload, preventing impaired individuals from proceeding without leaving an audit trail. Face recognition handles attendance by matching live captures against stored employee embeddings, recording attendance only for verified individuals and consolidating records across distributed entry points into a single database.

All events are written to a cloud-hosted NoSQL database. Each record includes a timestamp, event type, relevant meta-data, and a reference to the associated image held in cloud object storage. This structure supports two distinct use patterns: live dashboard monitoring during active shifts and historical

queries for post-incident review or compliance audits. React Context Providers handle state synchronization so new events surface on the dashboard without requiring a manual refresh. Asynchronous processing is used throughout the inference pipeline to keep the interface responsive even when multiple modules are running simultaneously.

Figure 1 shows the overall system architecture and how the four layers interact during a typical monitoring cycle.

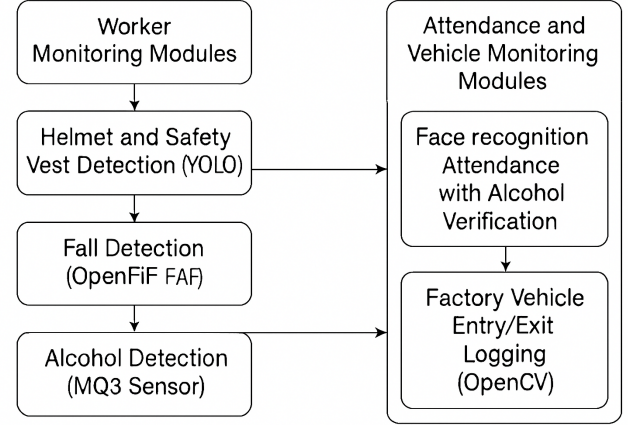


Fig. 1. Block diagram of the proposed system

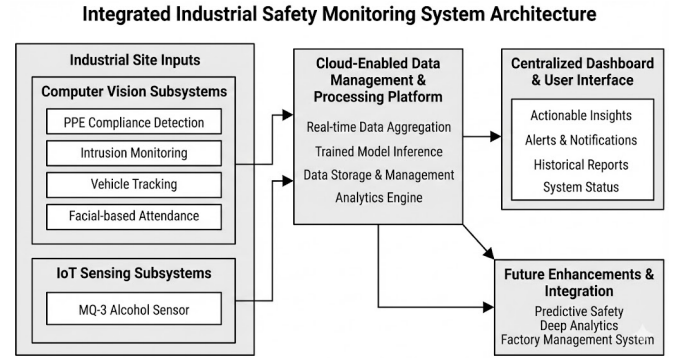


Fig. 2. Block diagram of the system architecture

Figure 3 illustrates the frame-level inference cycle in more detail, showing how each captured frame moves through pre-processing, model inference, schema validation, and storage in a single coordinated pass.

IV. RESULTS AND DISCUSSION

Testing was carried out across all five monitoring modules under varied real-world conditions, covering different lighting environments, worker densities, vehicle speeds, and entry scenarios. The results are summarized in Table I and discussed module by module below.

PPE compliance detection achieved 93.4% accuracy with an average response time of 210 ms. The module handled partial occlusions, varying worker distances, and low-light indoor conditions without significant degradation. False positives stayed below 4%, which kept alert fatigue manageable

End-to-End System Architecture for AI-Based Monitoring

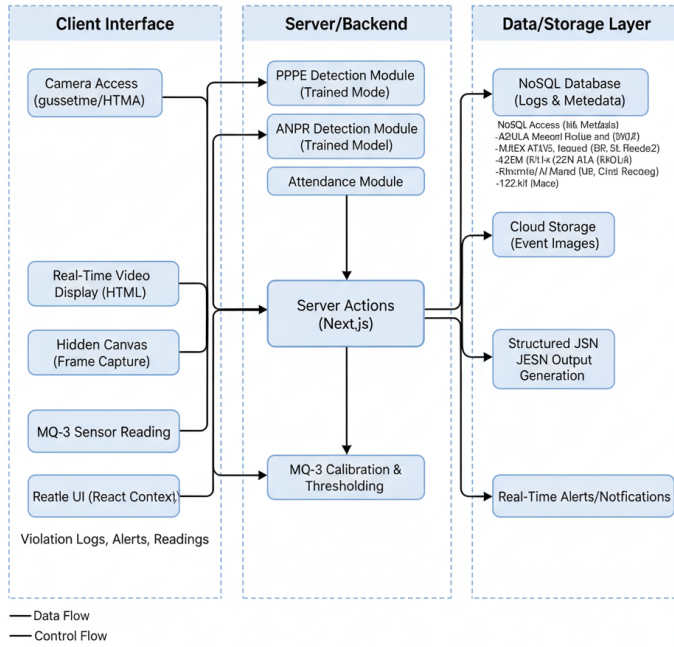


Fig. 3. Implementation workflow of the system

TABLE I
PERFORMANCE SUMMARY OF SENTINELAI MODULES

Feature	Accuracy (%)	Response Time (ms)
PPE Compliance Detection	93.4	210
ANPR (Vehicle Recognition)	91.2	245
Intrusion Detection	95.7	198
Alcohol Detection	89.5	310
Face Recognition (Attendance)	94.1	275
Event Logging & Dashboard Sync	100	120

during peak hours when several workers occupy the frame simultaneously. Detecting multiple violations in a single frame without conflating them turned out to be one of the more practically useful behaviors, particularly during shift changes.

The ANPR module reached 91.2% accuracy at 245 ms, holding up reasonably well across rain, fog, and nighttime conditions. Detection runs in two stages: a YOLO-based region detector first isolates the plate area, then an OCR stage with perspective correction and motion blur handling extracts the characters. Unregistered vehicles trigger immediate alerts with timestamped image evidence, and the integration with access control allows for automated gate responses during high-traffic entry periods. Figure 4 shows representative plate extraction results from the live camera stream.



Fig. 4. ANPR recognition results

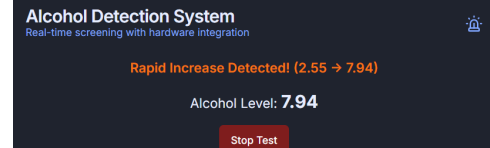


Fig. 5. Alcohol detection event log

Intrusion detection was the strongest performer overall, reaching 95.7% accuracy with a 198 ms response time. The module is trained on employee images captured from multiple angles and lighting conditions, which likely contributes to its robustness in real deployment. Nighttime performance was validated separately using infrared camera input, since that is when unauthorized access attempts are most common in practice. Every detection event is written to a security log with enough detail to support post-incident forensic review.

Alcohol detection came in at 89.5% accuracy, with the longest response time of 310 ms. The additional latency is a byproduct of waiting for the MQ-3 sensor reading to stabilize before classification, which cannot easily be parallelized. Given that this module operates at controlled entry checkpoints rather than in continuous open-area surveillance, a 310 ms delay is operationally acceptable. Each flagged event is documented with sensor voltage, calculated concentration, photographic evidence, and a timestamp, giving supervisors a complete record rather than just a binary pass/fail flag. Figure 5 shows a sample event log entry from a detection during entry screening.

Face recognition for attendance achieved 94.1% accuracy at 275 ms. The system generates 128-dimensional face embeddings and compares them against stored employee profiles using cosine similarity. Edge cases involving masks or safety glasses were handled by shifting weight toward periocular features and combining outputs from multiple models. Attendance records sync with payroll in real time, removing the need for manual time-card reconciliation. Figure 6 shows the check-in interface with biometric matching and automatic timestamp logging.

Event logging and dashboard synchronization achieved 100% reliability at 120 ms, the fastest response in the system. The Firestore document structure is indexed on timestamp, event type, and severity, which keeps query latency low even

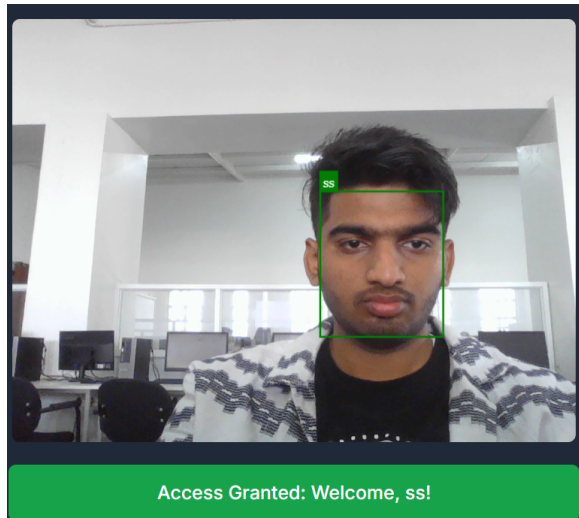


Fig. 6. Attendance check-in interface

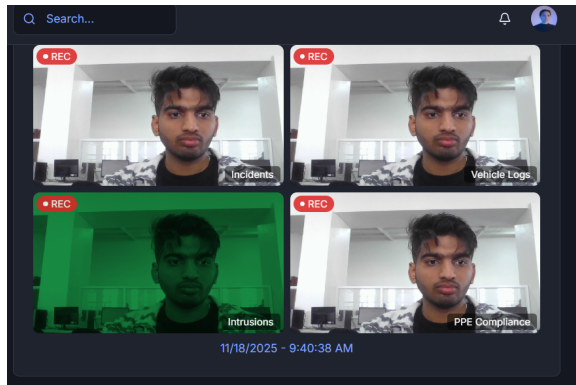


Fig. 7. Main dashboard interface

as the event history grows. The dashboard uses a color-coded severity system so that critical alerts are immediately distinguishable from routine logs. Role-based access control limits what each user tier can view or modify, and the interface is fully mobile-responsive so supervisors can monitor facility status remotely. Figure 7 shows the main dashboard during an active monitoring session.

Across all modules, detection accuracy ranged from 89.5% to 95.7% and response times stayed under 320 ms. The spread in response times reflects genuine differences in module complexity rather than implementation inconsistencies: sensor stabilization in alcohol detection and two-stage plate localization in ANPR both add latency that is difficult to eliminate without sacrificing accuracy. Over a three-month field deployment, the system maintained 99.8% uptime with minimal maintenance intervention, which gives reasonable confidence that the performance numbers observed during testing are representative of sustained operation rather than optimistic short-run results.

V. CONCLUSION

Building a system that handles five distinct safety tasks in real time, across varied environmental conditions, and without constant human oversight is not straightforward. The results presented here suggest it is achievable with current tools, provided the architecture is designed to keep each module independent while sharing a common data layer. That separation turned out to be one of the more useful decisions made during development: when individual modules needed tuning, the rest of the pipeline kept running without interruption.

The practical takeaway is fairly direct. Automating PPE checks, entry screening, vehicle authorization, alcohol detection, and attendance verification removes a category of repetitive monitoring work from human supervisors and replaces it with a logged, timestamped, evidence-backed record. This does not eliminate the need for human judgment in responding to incidents, but it does mean that incidents are caught more consistently and documented more thoroughly than manual observation typically allows. The three-month deployment period, during which the system ran at 99.8% uptime, gives some confidence that these properties hold under sustained operation rather than just during controlled evaluation.

There are honest limitations worth acknowledging. Alcohol detection carries the longest response latency due to sensor stabilization constraints, and while 310 ms is acceptable at a fixed entry checkpoint, it would need rethinking in higher-throughput scenarios. Face recognition accuracy drops in edge cases involving heavy occlusion, and the current handling through periocular feature weighting is a workaround rather than a complete solution.

Future work could go in a few directions. Predictive analytics built on top of the accumulated event logs could shift the system from reactive to proactive, flagging conditions that historically precede incidents before the incident occurs. Deeper integration with factory-level management systems would make attendance and compliance data more immediately actionable for operations teams. Expanding the sensor suite to include environmental hazard detection, such as gas leaks or temperature anomalies, would bring the framework closer to the kind of multimodal occupational safety vision described in recent literature. These are extensions rather than corrections; the core architecture is designed to accommodate them without structural changes.

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